Software testing case study

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Objective and scope

Check the behaviour of a small e-commerce journey and a separate product API. The website scenarios cover account states, sorting, cart changes and checkout information. The API scenarios cover retrieval, pagination, field selection, search, categories, sorting and two error responses.

SauceDemo provides the website exercise. Its problem_user account is an intentionally faulty training scenario. DummyJSON provides the separate API exercise; it was not tested as SauceDemo's backend.

Execution summary

Coverage	Result	Evidence
Website baseline / negative checks	7 passed	WEB-01 to WEB-07
Problem-account scenarios	3 failures reproduced	WEB-08 to WEB-10
Public API requests	10 passed; 55 assertions passed	API-01 to API-10

Approach

Expectations were taken from visible control labels, documented API behaviour and the standard_user comparison. Both valid and negative conditions were checked. The three website failures were retried in the same session; screenshots and field or cart state were retained with the results.

Environment and test data

Website: https://www.saucedemo.com | browser-assisted checks in an in-app browser on Windows.

API: https://dummyjson.com | Newman 6.2.3; HTTPS certificate verification enabled.

Accounts: Published demo accounts: standard_user, locked_out_user and problem_user.

Checkout data: QA / Tester / 12345; one Sauce Labs Backpack. Execution stopped at order review.

Execution window: 13:22:42-13:27:59 Cairo (UTC+3); API run 13:24:27-13:24:33.

Limits of the evidence

Coverage is limited to the recorded scenarios. Cross-browser behaviour, performance, completed orders and successful authenticated API calls were not tested. Exact browser engine version, viewport and deployed build were not captured. The original records are retained in the evidence folders.

Website test cases

WEB-01 to WEB-04 | SauceDemo

WEB-01 Sign in with a valid account

Setup: Open https://www.saucedemo.com/ with no account signed in.

Input: Username standard_user; use the published password shown on the login page.

Steps: 1. Enter standard_user in Username. 2. Enter the published demo password. 3. Select Login. 4. Inspect the destination and inventory.

Expected: The inventory page opens and the product catalog is visible.

Observed: The browser reached /inventory.html. All six catalog products were visible, with distinct product-image sources. These images supplied the normal inventory reference for the later problem_user comparison.

Result: PASS

WEB-02 Reject a locked account

Setup: Open the demo login page with no account signed in.

Input: Username locked_out_user; use the published demo password.

Steps: 1. Enter locked_out_user in Username. 2. Enter the published demo password. 3. Select Login. 4. Inspect the error message and current page.

Expected: Login is refused with a locked-account message; the inventory does not open.

Observed: The page displayed “Epic sadface: Sorry, this user has been locked out.” The browser remained at https://www.saucedemo.com/. No inventory page was reached by the rejected login attempt.

Result: PASS

WEB-03 Sort products by price

Setup: Sign in as standard_user and open inventory. All six catalog products are present.

Input: Sort option: Price (low to high).

Steps: 1. Open the product sort selector. 2. Select Price (low to high). 3. Read every displayed price from the first product to the last. 4. Check that all six products remain visible.

Expected: All six products remain present and prices appear in non-decreasing numerical order. Equal prices are valid.

Observed: Prices were $7.99, $9.99, $15.99, $15.99, $29.99 and $49.99. All six products remained present.

Result: PASS

WEB-04 Add a backpack to the cart

Setup: Sign in as standard_user; open inventory with an empty cart. Remove any existing items before starting.

Input: Product: Sauce Labs Backpack; listed price $29.99.

Steps: 1. Find Sauce Labs Backpack. 2. Select its Add to cart button. 3. Open the cart. 4. Inspect the product name, quantity, price and available control.

Expected: The cart contains one Sauce Labs Backpack, quantity 1, priced at $29.99, with a Remove control.

Observed: The cart displayed the backpack, quantity 1, $29.99 and a Remove control.

Result: PASS

Website test cases

WEB-05 to WEB-07 | SauceDemo

WEB-05 Remove the only cart item

Setup: Continue from WEB-04 as standard_user. The cart contains only one Sauce Labs Backpack.

Input: Remove control for Sauce Labs Backpack.

Steps: 1. Open the cart if it is not already displayed. 2. Select Remove for the backpack. 3. Inspect the remaining product rows and navigation controls.

Expected: The backpack disappears and no product rows remain. Cart navigation remains available.

Observed: No product row remained after removal. The cart still displayed its headings, Continue Shopping and Checkout controls.

Result: PASS

WEB-06 Reject checkout with empty information

Setup: Sign in as standard_user. Add one Sauce Labs Backpack to an otherwise empty cart; open Cart and select Checkout.

Input: First Name, Last Name and Zip/Postal Code: all empty.

Steps: 1. Verify that all three information fields are empty. 2. Select Continue. 3. Inspect the validation message and page.

Expected: Checkout remains on the information page and identifies First Name as required.

Observed: “Error: First Name is required” appeared; the URL remained /checkout-step-one.html. Separate validation of Last Name and postal code was not tested by this case.

Result: PASS

WEB-07 Continue with valid checkout information

Setup: Sign in as standard_user. Keep one Sauce Labs Backpack in the cart; open Cart and select Checkout.

Input: First Name QA; Last Name Tester; Zip/Postal Code 12345. All data is invented.

Steps: 1. Enter QA in First Name. 2. Enter Tester in Last Name. 3. Enter 12345 in Zip/Postal Code. 4. Inspect the three values. 5. Select Continue.

Expected: Values remain in their own fields and checkout advances to the overview.

Observed: The fields retained QA / Tester / 12345. The overview opened at /checkout-step-two.html with the backpack and total $32.39. Finish was not selected.

Result: PASS

Website test cases

WEB-08 to WEB-10 | SauceDemo

FAIL denotes a failed shopper-facing expectation on the deliberate problem_user scenario; it also confirms that the seeded behaviour was reproduced.

WEB-08 Check catalog product photographs

Setup: Sign in as problem_user and open inventory. Use the standard_user inventory from WEB-01 as the comparison.

Input: All six product photographs.

Steps: 1. Inspect each loaded product photograph. 2. Compare each photograph and image source with the standard_user baseline. 3. Reload inventory. 4. Repeat the inspection.

Expected: Each named product displays its corresponding photograph, consistent with the baseline.

Observed: All six loaded images showed the same dog photograph and shared one source. The outcome persisted after reload: 2/2 observations in one session. A standard_user control confirmed six distinct, loaded product images.

Result: FAIL - seeded defect reproduced; LIVE-BUG-01

WEB-09 Add a second product to the cart

Setup: Prepare one backpack in the cart as standard_user, then log out and sign in as problem_user. Open inventory; Bolt T-Shirt is absent; cart count is 1.

Input: Sauce Labs Bolt T-Shirt; Add to cart.

Steps: 1. Select Add to cart for Bolt T-Shirt. 2. Inspect its button and cart count. 3. Reload inventory. 4. Repeat the click. 5. Open Cart and inspect the product rows.

Expected: The T-Shirt appears in the cart, its button becomes Remove and count increases to 2.

Observed: Both attempts left Add to cart and count 1. Cart contained only the backpack. A same-item standard_user control showed Remove and count 2.

Result: FAIL - seeded defect reproduced; LIVE-BUG-02

WEB-10 Retain Last Name during checkout

Setup: Prepare one backpack as standard_user, then log out and sign in as problem_user. Open Cart and select Checkout.

Input: QA / Tester / 12345, as in WEB-07.

Steps: 1. Enter QA in First Name. 2. Enter Tester in Last Name. 3. Enter 12345 in Zip/Postal Code. 4. Inspect each field value. 5. Select Continue. 6. Reload the information page. 7. Repeat using sequential keystrokes.

Expected: Each field retains its own value; checkout advances to the overview.

Observed: Last Name stayed empty. Browser field filling changed First Name to Tester; sequential typing left it as r. Both attempts showed “Error: Last Name is required” and prevented progression. The standard_user baseline accepted identical data.

Result: FAIL - seeded defect reproduced; LIVE-BUG-03

Defect report

Last Name entry changes First Name and blocks checkout

Reference	Severity	Proposed priority
LIVE-BUG-03 | WEB-10	High	P1

Reproduction

Setup: Prepare one backpack in the cart as standard_user. Log out and sign in as problem_user. Open Cart and select Checkout. Use invented data QA / Tester / 12345.

1. Enter QA in First Name.
2. Enter Tester in Last Name.
3. Enter 12345 in Zip/Postal Code.
4. Inspect the three field values.
5. Select Continue.
6. Reload the information page.
7. Repeat steps 1-5 using sequential keystrokes.

Expected: First Name retains QA, Last Name retains Tester and postal code retains 12345. Continue opens the checkout overview.

Observed: Last Name remained empty. Browser field filling changed First Name to Tester; sequential typing left First Name as r. Postal code retained 12345. Both attempts displayed “Error: Last Name is required” and did not advance.

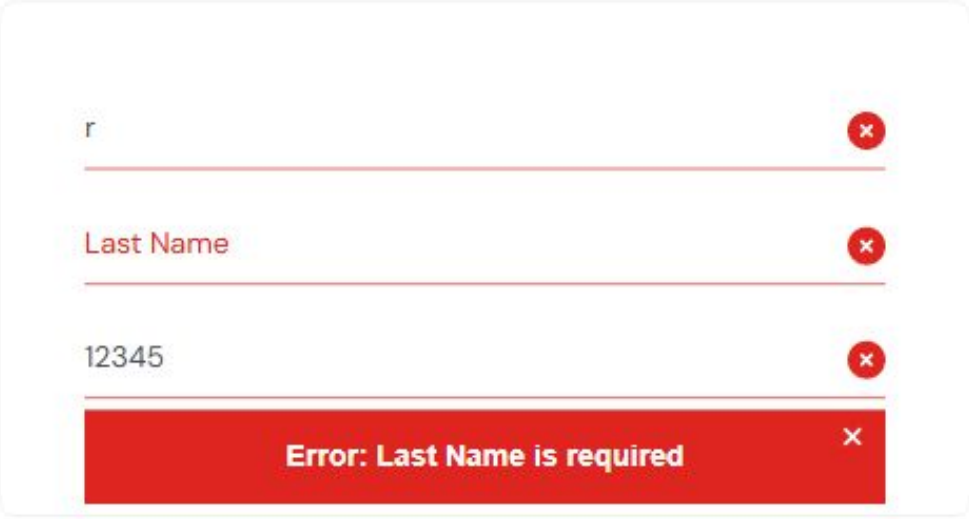
Impact: The problem_user account cannot complete the required information, blocking checkout before order review.

Repeat check: 2/2 attempts in one session: one browser field-fill attempt and one post-reload sequential-keystroke attempt.

Control comparison: WEB-07 accepted identical data for standard_user and reached the checkout overview.

Priority rationale: Address first. Valid input cannot satisfy a required field, preventing the tested account from reaching order review.

Evidence detail



Last Name remains empty; First Name contains r; required-field validation blocks Continue. Detail from 15-problem-checkout-reproduced.png.

Evidence: 13-problem-checkout-fields.png, 14-problem-checkout-blocked.png, 15-problem-checkout-reproduced.png, 06-standard-checkout-fields.png, 07-standard-checkout-overview.png. Original files: Evidence/Browser/.

Defect report

Bolt T-Shirt is not added to the cart

Reference	Severity	Proposed priority
LIVE-BUG-02 | WEB-09	High	P2

Reproduction

Setup: Prepare a cart containing one Sauce Labs Backpack as standard_user. Log out and sign in as problem_user. Open inventory; cart count is 1 and Bolt T-Shirt shows Add to cart.

1. Select Add to cart for Sauce Labs Bolt T-Shirt.
2. Inspect its button and the cart count.
3. Reload inventory.
4. Select the same Add to cart button again.
5. Open Cart.
6. Inspect the product rows.

Expected: The T-Shirt is added alongside the backpack, its control becomes Remove and cart count increases from 1 to 2.

Observed: Both clicks left Add to cart visible and count at 1. No Remove control appeared for this product. The cart contained only the backpack, confirming that this was not merely an incorrect badge.

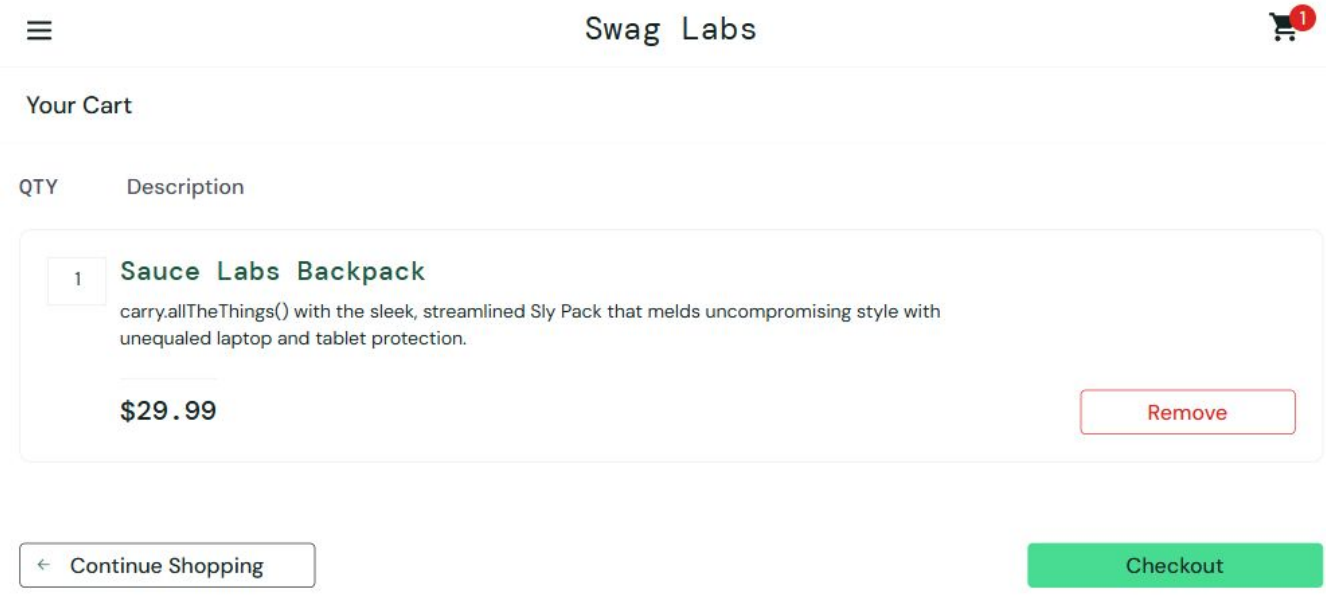
Impact: The problem_user account cannot add the T-Shirt for purchase through its Add to cart control.

Repeat check: 2/2 click attempts in one session; the second followed a reload.

Control comparison: The same-item standard_user control showed Remove and cart count 2.

Priority rationale: Schedule after the checkout blocker. A core cart action fails, but the confirmed scope is one product/account and the backpack remains available.

Evidence detail



After the Add to cart attempts, only the backpack is present. The same-item standard_user control is retained separately.

Evidence: 10-problem-add-to-cart-first.png, 11-problem-add-to-cart-reproduced.png, 12-problem-cart-item-absent.png, 17-standard-bolt-add-control.png. Original files: Evidence/Browser/.

Defect report

All six products display the same incorrect photo

Reference	Severity	Proposed priority
LIVE-BUG-01 | WEB-08	Medium	P2

Reproduction

Setup: Open SauceDemo inventory as problem_user using the published demo credentials. Retain the standard_user inventory screenshot from WEB-01 as a comparison.

1. Inspect the photograph for each of the six named products.

2. Compare the photographs and image sources with the standard_user baseline.

3. Reload inventory.

4. Inspect the six photographs again.

Expected: Every product displays its corresponding photograph. The standard_user comparison shows six distinct product-image sources.

Observed: All six products displayed the same dog photograph. Each image was fully loaded and used /assets/sl-404-Cq1a9k9X.jpg. The same result remained after reload; this was incorrect loaded content rather than an incomplete image load.

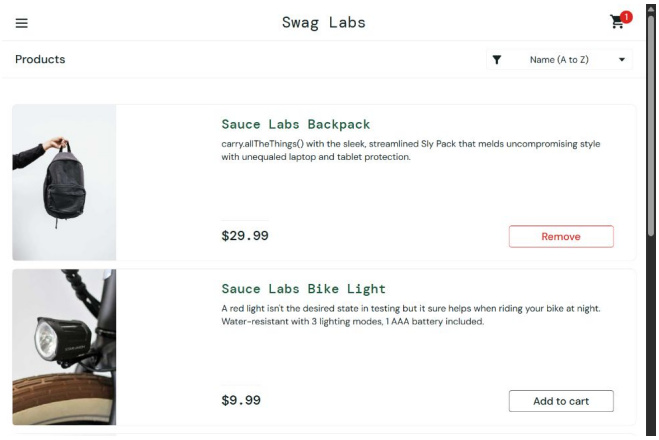
Impact: Incorrect photos make the products harder to recognise. Names and prices remain visible, so customers can still identify items by text.

Repeat check: 2/2 observations in one session: initial inspection and one inspection after reload. Coverage does not include another browser or device.

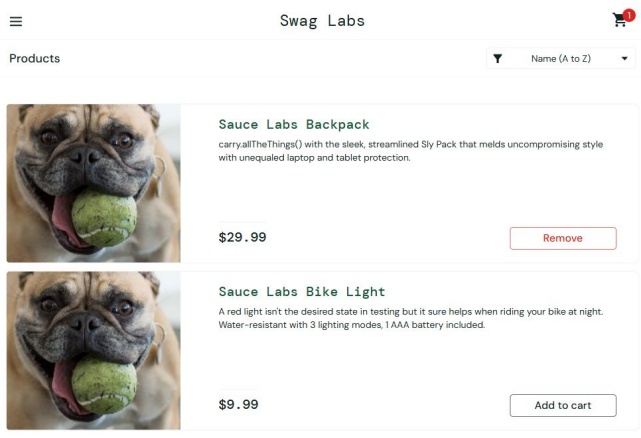
Control comparison: The standard_user control confirmed six loaded, distinct product images.

Priority rationale: Schedule after the checkout blocker. Product recognition is affected across the catalog, while names, prices and cart controls remain available.

Control comparison



standard_user: product photographs



problem_user: repeated dog photograph

Evidence: 09-problem-product-images.png, 16-standard-image-control.png, 01-standard-inventory.png. Original files: Evidence/Browser/.

API validation

DummyJSON | 10 read-only requests | 55 assertions passed

The collection checks response status, JSON content, data shape and field values. The two negative cases distinguish an unavailable product from a token-protected endpoint requested without credentials. Response times are retained as diagnostic data; no performance threshold was asserted.

Case	Scenario	HTTP	Checks
API-01	Existing product	200	6/6 pass
API-02	Page size and selected fields	200	6/6 pass
API-03	Page offset	200	6/6 pass
API-04	Keyword search	200	6/6 pass
API-05	No-match search	200	5/5 pass
API-06	Category filter	200	6/6 pass
API-07	Ascending price sort	200	6/6 pass
API-08	Category-name list	200	6/6 pass
API-09	Missing product	404	4/4 pass
API-10	Missing access token	401	4/4 pass

Repeat the collection

1. Import API/Live_DummyJSON.postman_collection.json into Postman.

2. Keep baseUrl as https://dummyjson.com. Use No Auth and one iteration.

3. Clear existing dummyjson.com authentication cookies before API-10, which requires an unauthenticated request.

4. Run all ten requests and inspect both response bodies and assertion results. Fixture changes or transport errors should be investigated before raising a defect.

Traceability and references

03_Web_Test_Cases.md and 04_Defect_Reports.md contain complete, editable records and evidence links. API/API_Test_Cases.md contains API inputs, expectations and actual results. The Evidence folder preserves the original screenshots, JSON records and collection fingerprint.

UI test basis: visible labels and standard_user controls. API test basis: official documentation and the fixture assumptions stated in each case. Priorities in this report use P1 for the checkout blocker and P2 for the other findings. These are proposed case-study values.

Sauce Labs sample application

DummyJSON product API

DummyJSON authentication